



COMSATS University Islamabad, Lahore Campus
Department of Electrical and Computer Engineering

Mid Term Examination – FALL 2021

Course Title:	Digital Image Processing	Course Code:	CPE415	Credit Hours:	4(3,1)
Course	Dr. Ikramullah Khosa	Program Name:	BE Computer Engineering		
Semester:	7 th	Batch:	FA18-BCE	Section:	A/B
Date:	15-11-2021				
Time Allowed:	90 Minutes			Maximum Marks:	25
Student's Name:			Reg. No.	CIIT/DDP- /LHR	
Important Instructions / Guidelines:					
This is a closed-book, closed-notes examination.					

Question 1 (CLO-1/ PLO-1/C3) (10 Marks)

Apply the concepts of image fundamentals and image enhancement to solve the following questions.

- Estimate the memory required in MB (mega bytes) for storing an HD color video recorded at 60 frames per second of 3 minutes duration? (Hint: HD resolution = 1920×1080) (1 Marks)
- A CCD camera chip of dimension 14×14 mm, and having 4096×4096 elements is focused on a square, flat area, located 0.5m away. How many line pair per mm will this camera be able to resolve? The camera is equipped with a 35 mm lens. (2 Marks)
- Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass. (3 Marks)
- Consider the following segment of 3-bit image. Show the binary equivalent of the image, as well as the individual bit planes. (4 Marks)

5	1	3	0	2
0	4	1	5	1
3	2	7	2	3
6	3	0	6	2
2	1	7	3	4

Question 2 (CLO-1/ PLO-1/C3) (5 Marks)

Apply the concepts of image filtering in the frequency domain to solve the following questions.

Consider a 1-dimensional 1×5 spatial mask estimates the average of four horizontally closest neighbors of a point (x,y), but excludes the point itself from the average. Find the equivalent filter H(u,v) in the frequency domain. Also show that your result is a low pass filter. (3+2 Marks)

Question 3 (CLO-2/ PLO-2/C4) (10 Marks)

Apply the concepts of image restoration to solve the following questions.

- Analyze the expression for expected outcome of optimum notch filter shown in Eqn. (1). Outline the expression for weights $w(x,y)$ to be used in optimum notch filter to reject the multi-spectral noise. (4 Marks)

$$\hat{f}(x,y) = g(x,y) - w(x,y)\eta(x,y) \quad (1)$$

- Illustrate and label six different noise distributions which may affect a digital image. (6 Marks)



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Terminal Examination – FALL 2021

Final Examination – FALL 2021						
Course Title:	Digital Image Processing			Course Code:	CPE415	Credit Hours: 4(3,1)
Course	Dr. Ikramullah Khosa			Program Name:	BS Computer Engineering	
Semester:	7 th	Batch:	FA18-BCE	Section:	A/B	Date: 10-01-2022
Time Allowed:	3 Hours			Maximum Marks:		50
Student's Name:	Muhammad Naimat Ali			Reg. No.	FA18-BCE-070-A	
<u>Important Instructions / Guidelines:</u>						
This is a closed-book, closed-notes examination.						

Question 1 (CLO-1/PLO-1)

(7 Marks)

The probabilities of individual gray levels for a 3-bit coded image are given in the following table. Apply histogram equalization and **compute** the new pixel intensities after equalization. Show the original and equalized histograms as well as the transformation function.

r_k	$P_r(r_k) = n_k/MN$
$r_0 = 0$	0.2
$r_1 = 1$	0.3
$r_2 = 2$	0.25
$r_3 = 3$	0.15
$r_4 = 4$	0.1
$r_5 = 5$	0
$r_6 = 6$	0
$r_7 = 7$	0

Question 2 (CLO-1/PLO-1)

(7 Marks)

Consider the image shown in Figure 1. Apply the concepts of color image processing to show the CMY components of the image as they would appear on the monochrome monitor.

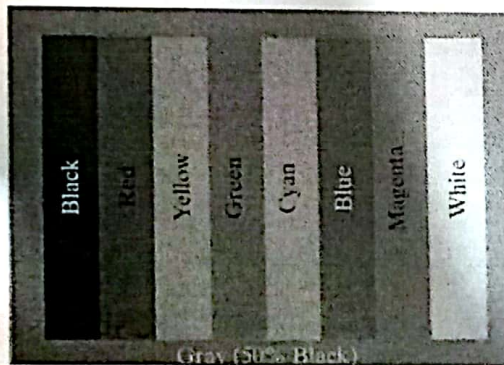


Figure 1

Question 3 (CLO-2/PLO-2)

(7 Marks)

Relate the concepts of image segmentation and **identify** that the Sobel and Prewitt masks give isotropic results only for horizontal and vertical edges, and for edges oriented at $\pm 45^\circ$ respectively.

Question 4 (CLO-2/PLO-2)

(7 Marks)

Relate the concepts of image segmentation to **identify** whether the threshold obtained with the basic global thresholding algorithm is independent of the starting point? If your answer is yes, prove it. If your answer is no, give an example.

Question 5 (CLO-2/PLO-2)**(7 Marks)**

Analyze the image shown in Figure 2 and deconstruct into segments using split and merge procedure (employing Quad Tree method). Show the final quad tree after the segmentation process is finished.

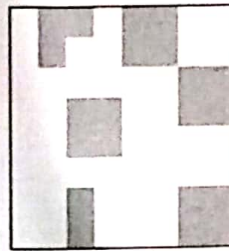


Figure 2

Question 6 (CLO-2/PLO-2)**(7 Marks)**

Analyze the shape shown in the Figure 3 and outline the following:

- 1) Chain Code (using general 8-directional code) with the order of 12 (i.e. 12 segments) (2 Mark)
- 2) Difference (2 Mark)
- 3) Shape number (1 Mark)
- 4) Signature (2 Mark)

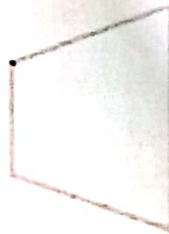


Figure 3

Question 7 (CLO-2/PLO-2)**(8 Marks)**

Analyze the image shown in Figure 4 and illustrate the two Gray Level Co-occurrence Matrices (GLCMs). To identify the size of the matrix, consider the pixel intensity range of the image. The position operators to calculate two GLCMs are:

- 1) One pixel to right and one pixel below (4 Mark)
- 2) One pixel to left and two pixels below (4 Mark)

2	1	2	4
4	1	2	2
1	5	4	3
5	4	5	1
2	3	1	3

Figure 4

End of paper